UNDE ELECTROMAGNETICE

1. Ec. MAXWELL sub formă integrală și locală
2. Ec. dif. a undelor electromagnetice. Soluție: undă plană armonică
3. Pp. undelor elem. plane
4. Energia undei. Vectorul lui POYNTING. Intensitatea undei.

---

1. integrale:

a) $\Phi_e = \oint\!\!\oint_S \vec{E}\cdot d\vec{S} = \frac{q_{int}}{\varepsilon_0} = \frac{1}{\varepsilon_0}\iiint_V \rho\, dV$ — Th. Gauss

- $\oint\!\!\oint_S$: integrală pe suprafață închisă.
- $\vec{E}\cdot d\vec{S}$: p. scalar ($\cos\alpha$)
- q_{int}: sarcina totală din volumul închis de suprafața considerată

b) $e_{em} = \oint_\Gamma \vec{E}\cdot d\vec{l} = -\frac{d\Phi_m}{dt} = -\iint_S \frac{\partial \vec{B}}{\partial t}\cdot d\vec{S}$ — legea lui Faraday

- $\oint_\Gamma$: contur închis (imaginar)
- $\iint_S$: suprafața care se sprijină pe conturul dat

c) $\Phi_m = \oint\!\!\oint \vec{B}\cdot d\vec{S} = 0$

d) $\oint \vec{B}\cdot d\vec{l} = \mu_0 (I + I_d)$, $I = \iint \vec{j}\cdot d\vec{S}$ → ... — legea Maxwell-Ampère.

Sub formă locală:

$$\begin{cases} \operatorname{div}\vec{E} = \frac{\rho}{\varepsilon_0} & (\nabla\cdot) \\ \operatorname{rot}\vec{E} = -\frac{\partial \vec{B}}{\partial t} & (\nabla\times) \\ \operatorname{div}\vec{B} = 0 \\ \operatorname{rot}\vec{B} = \mu_0\vec{j} + \varepsilon_0\mu_0\frac{\partial \vec{E}}{\partial t} \end{cases}$$

$\vec{E}(t) \longrightarrow \vec{B}(t)$

$\vec{D} = \varepsilon_0\varepsilon_r\vec{E}$; $\vec{B} = \mu_0\mu_r\vec{H}$; $\vec{j} = \sigma\vec{E}$

pt vid.: $\vec{j}=0$, $\rho=0$

$$\begin{cases} \nabla\cdot\vec{E}=0 \\ \nabla\cdot\vec{B}=0 \\ \nabla\times\vec{E}=-\dfrac{\partial\vec{B}}{\partial t} \\ \nabla\times\vec{B}=\varepsilon_0\mu_0\dfrac{\partial\vec{E}}{\partial t} \end{cases}$$

(M_1) $\nabla\cdot\vec{E}=0$

(M_2) $\nabla\cdot\vec{H}=0$

(M_3) $\nabla\times\vec{E}=-\mu_0\dfrac{\partial\vec{H}}{\partial t}$

(M_4) $\nabla\times\vec{H}=\varepsilon_0\dfrac{\partial\vec{E}}{\partial t}$

$\vec{E}(\vec{r},t)$? $\vec{B}(\vec{r},t)$?

folosim identitatea matem.:

$$\nabla\times(\nabla\times\vec{A})=\nabla(\nabla\cdot\vec{A})-\nabla^2\vec{A}$$

$$(\text{rot}(\text{rot}\,\cdot)) = \text{grad}(\text{div}\,\cdot)-\Delta$$

Aplicăm $\nabla\times(M_3)\Rightarrow \nabla\times(\nabla\times\vec{E})=\nabla\times\left(-\dfrac{\partial\vec{B}}{\partial t}\right)$

$\nabla\times(\nabla\times\vec{E})$ —(def.)→ $\nabla(\nabla\cdot\vec{E})-\nabla^2\vec{E}$ —(M_1)→ 0

$\nabla\times\left(-\dfrac{\partial\vec{B}}{\partial t}\right)=-\dfrac{\partial}{\partial t}(\nabla\times\vec{B})$ —(M_4)→ $-\varepsilon_0\mu_0\dfrac{\partial\vec{E}}{\partial t}$

$$-\nabla^2\vec{E}=-\varepsilon_0\mu_0\frac{\partial^2\vec{E}}{\partial t^2}$$

$$\boxed{\Delta\vec{E}-\varepsilon_0\mu_0\frac{\partial^2\vec{E}}{\partial t^2}=0}$$

ec. dif. a unei unde

analog: ……>

$$\boxed{\Delta\vec{B}-\varepsilon_0\mu_0\frac{\partial^2\vec{B}}{\partial t^2}=0}$$

$$\boxed{\Delta\Psi-\frac{1}{v^2}\cdot\frac{\partial^2\Psi}{\partial t^2}=0}$$ → ec. dif. a unei: ;

$\Psi \leftrightarrow \vec{E}, \vec{B}$

prin identificare $\Rightarrow$ $\boxed{v=\dfrac{1}{\sqrt{\varepsilon_0\mu_0}}}$

Soluția: unda plană: $\vec{\Psi}(\vec{r},t)=\vec{A}e^{i(\omega t-\vec{k}\cdot\vec{r})}$, $\vec{k}\cdot\vec{r}=k_x x+k_y y+k_z z$

sau: $\vec{\Psi}(\vec{r},t)=\vec{A}\cos(\omega t-\vec{k}\cdot\vec{r})$

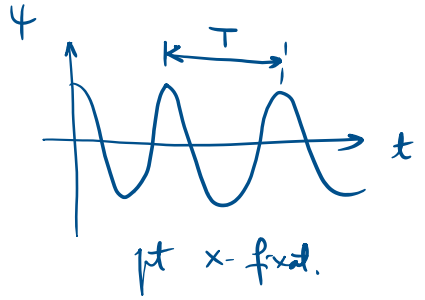
$\varphi(\vec{r},t) = \omega t - \vec{k}\cdot\vec{r}$ — faza undei

1D: Ox : $\varphi(x,t) = \omega t - kx$.

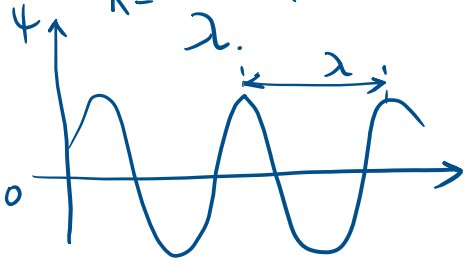
$(\omega; k)$ ⇓ $(T; \lambda)$.

$\omega = \frac{2\pi}{T}$ (pulsatia)

$k = \frac{2\pi}{\lambda}$ (nr. de unda)



ψ pt x - fixat.



ψ pt t - fixat

viteza undei (suprafetele de faza constanta)

$d\varphi = 0 \Rightarrow \omega\,dt - k\,dx = 0$

$\Rightarrow \boxed{v = \frac{dx}{dt} = \frac{\omega}{k} = \frac{\lambda}{T}}$.

în vid:

$v = \frac{1}{\sqrt{\varepsilon_0\mu_0}} = \frac{1}{\sqrt{8{,}85\ldots}} = 3\cdot 10^8$ m/s !! = (c)

LUMINA = undă electromagnetică

pt. un mediu oarecare:

$v = \frac{1}{\sqrt{\varepsilon\mu}} = \frac{1}{\sqrt{\varepsilon_0\varepsilon_r\mu_0\mu_r}} = \frac{1}{\sqrt{\varepsilon_0\mu_0}} \cdot \frac{1}{\sqrt{\varepsilon_r\mu_r}} = \frac{c}{n}$.

n — indice de refracție!

(3) Proprietăți: pt $\vec{E} = \vec{E}_0 e^{i(\omega t - \vec{k}\cdot\vec{r})}$ → $k_x x + k_y y + k_z z$

$\nabla\cdot\vec{E} \longrightarrow -i\vec{k}\cdot\vec{E}$

$\nabla = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$

$\nabla\cdot \longrightarrow -i\vec{k}\cdot$

$\nabla\times \longrightarrow -i\vec{k}\times$

$\frac{\partial}{\partial t} \longrightarrow i\omega$

(A): $\vec{E} = \vec{E}_0 \, e^{i(\omega t - \vec{k}\cdot\vec{r})}$

$\vec{B} = \vec{B}_0 \, e^{i(\omega t - \vec{k}\cdot\vec{r})}$

Dacă pt. propagare pe axa $\underline{Oz}$:

$\vec{E} = \vec{E}_0 \cos(\omega t - kz)$

$\vec{B} = \vec{B}_0 \cos(\omega t - kz)$

$\Downarrow$

$\vec{E}$ și $\vec{B}$ sunt în FAZĂ

(trec simultan prin max. și minime)

(B) din $(M_1) \Rightarrow \nabla \cdot \vec{E} = 0 \longrightarrow -i\vec{k}\cdot\vec{E} = 0 \Rightarrow \vec{k} \perp \vec{E}$

$(M_2) \Rightarrow \nabla \cdot \vec{B} = 0 \quad \Rightarrow \vec{k} \perp \vec{B}$

undă transversală

(C) din $(M_3) \Rightarrow \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$\downarrow \qquad \downarrow$

$-i\vec{k} \times \vec{E} = -i\omega \vec{B}$

$\Rightarrow \vec{k} \times \vec{E} = \omega \vec{B} \Rightarrow \begin{cases} \vec{B} \perp (\vec{k}, \vec{E}) \\ kE = \omega B \end{cases}$

din $(M_4) \Rightarrow \nabla \times \vec{B} = \varepsilon\mu \frac{\partial \vec{E}}{\partial t} \Rightarrow$

$\downarrow$

$-i\vec{k} \times \vec{B} = \varepsilon_0 \mu_0 i\omega \vec{E} \Rightarrow -\vec{k} \times \vec{B} = \varepsilon_0 \mu_0 \omega \vec{E}$

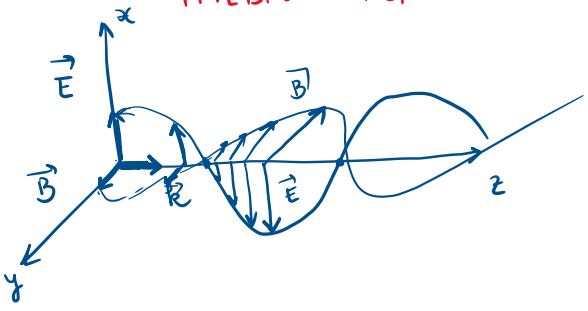
$\Downarrow$

$\begin{cases} \vec{E} \perp (\vec{B}, \vec{k}) \\ kB = \varepsilon_0 \mu_0 \omega E \end{cases}$

$\vec{B} \times \vec{k}$

$(\vec{E}, \vec{B}, \vec{k})$

TRIEDRU DREPT.



(D). Legătura între E și B (valori)

$$kE = \omega B \Rightarrow E = \frac{\omega}{k} B = \frac{\frac{2\pi}{T}}{\frac{2\pi}{\lambda}} B = \left(\frac{\lambda}{T}\right) B.$$

în vid: $\boxed{E = cB}$.

în $\underline{n}$: $E = vB$.

$$E = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} \cdot \mu_0 H \Rightarrow E = \sqrt{\frac{\mu_0}{\varepsilon_0}} H \Rightarrow \boxed{\sqrt{\varepsilon_0} E = \sqrt{\mu_0} H}.$$

---

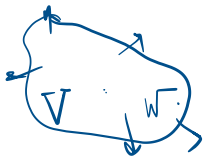
4). $w = w_e + w_m = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\mu_0 H^2 = \frac{1}{2}\vec{E}\cdot\vec{D} + \frac{1}{2}\vec{H}\cdot\vec{B}$

densit. volumică de energie

$\vec{E}\,\varepsilon_0\vec{E} \rightarrow \vec{D}$

$$W = \iiint w \, dV$$

elem. de volum



th. de conservare: $-\frac{dW}{dt} = \oiint (\vec{E} \times \vec{H}) \cdot d\vec{S}$ → elem. de suprafață!

$\vec{S}$ (vector POYNTING)

(direcția de propagare a energiei)

Def:

Intensitatea undei:

$$I = \frac{1}{T} \cdot \int_0^T S(t)\,dt = \frac{1}{T} \sqrt{\frac{\varepsilon_0}{\mu_0}} E_0^2 \int_0^T \cos^2(\omega t - kx)\,dt$$

$\frac{1}{2}T$

$$= \frac{1}{2}\sqrt{\frac{\varepsilon_0}{\mu_0}} E_0^2$$

$$= \frac{1}{2}\sqrt{\frac{\mu_0}{\varepsilon_0}} H_0^2.$$